

EDUCATION

Degree	Institute/Board	CGPA/Percentage	Year
BS Software Engineering	FAST-NUCES, Peshawar	2.52/4 (Current)	2022–Present
Intermediate	BISE, Mirpurkhas	80.72%	2020
Matriculation	BISE, Mirpurkhas	80.28%	2018

WORK EXPERIENCE

- FREELANCER

2020 - Present

Web Development

Online

– Developed and maintained **custom websites** for clients across diverse industries, delivering projects on schedule.

– Implemented **responsive, user-friendly designs** using HTML, CSS, and JavaScript.

– Provided **post-launch support and troubleshooting**, addressing issues and implementing improvements.

– Documented **code and project updates**, creating comprehensive notes and progress reports for clients.

PROJECTS

- AGENT BASED VISION-MIXER FOR VIDEO BROADCASTS (Final Year Project)

Aug 2025 – Present

Supervisor: Dr. Omar Usman Khan | FAST NUCES

– Built an **intelligent multi-camera vehicle surveillance system** using **YOLOv8** for real-time detection and **DeepSORT** for single-camera tracking across parallel RTSP streams.

– Implemented **cross-camera vehicle re-identification** using **ResNet-IBN** appearance embeddings and cosine similarity, assigning persistent global IDs across the entire camera network.

– Designed a **Dynamic Bayesian Network (DBN)** scoring engine that automatically switches the active feed to the best-quality camera using hysteresis and frame-buffer stability checks.

– Automated **video stitching** of multi-camera journey clips via **FFmpeg** and delivered compiled footage to recipients via **SMTP email**, with a full desktop GUI for live monitoring.
- HOTEL MANAGEMENT SYSTEM

Sec 2023 – Nov 2023, FAST

Guide: Usman Wajid

Github

– Demonstrated **Object-Oriented Programming (OOP)** principles in **C++** with features like room booking, billing, and etc. Focused on modular design, data persistence, and code reusability.
- CONNECT-SPHERE - COMMUNITY ENGAGEMENT APP

Feb 2024 - Apr 2024, FAST

Guide: Muhammad Umer Haroon

Github

– Built using **Flask** and **Tkinter** to connect users and enable community interaction with real-time data exchange.

– Integrated **Firebase** for authentication and **secure data storage**.
- OTHER PROJECTS

Aug 2024 - Sep 2025

Self Guided / FAST

– **AI-Driven Matrimonial Platform**, PHP web app with Flask API, SQL database, ML matchmaking, and chatbot.

– **Pathfinding Algorithm Visualizer**, Python tool visualizing BFS, Dijkstra, Bellman-Ford, Floyd-Warshall.

– **Multi-Agent Stock Trading System**, AI simulation using Naive Bayes, K-Means, and multi-agent framework.

– **Weather App**, Python app fetching real-time weather via Weather API.

TECHNICAL SKILLS

- Programming Languages: C/C++, Python, Java, SQL, HTML , CSS, JavaScript, PHP

• Software/Tools/libraries/Frameworks: Visual Studio, Git, Ubuntu, Flask, FastAPI, Bootstrap, Laravel, Jupyter Notebook, Pycharm, Wireshark, IntelliJ IDEA, Cisco Packet Tracer, Anylogic, Jira, Trello, ProjectLibre, Docker & Kubernetes

CERTIFICATIONS

- Uwrox - Self Alignment to Startup,Focused on startup alignment program and community services.

02/2023
- CodSoft (Internship) - Programming in C++,Completed an internship covering all advanced c++ concepts.

12/2023
- Microsoft - Front-End Development, Completed a Microsoft-certified course.

01/2024
- Coursera - C# Course Completion, Successfully completed a C# programming course on Coursera.

10/2025
- Google - Gemini Certified Student, Earned certification as a Gemini Certified Student by Google.

10/2025
- CISCO Networking Academy - Data Science Course, Completed the Data Science course.

11/2025